MEAN

Case 3: Test of significance of the difference between sample mean and population mean.

$\overline{x}$ - Sample Mean, n - Sample Size, μ - Population Mean and σ - Population Standard Deviation (S.D.)

$$\text{Test Statistic } z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Note: If σ is not known, the sample S.D. 's' can be used as 's' tends to σ when n is sufficiently large.

A sample of 900 members has a mean 3.4 cm and S.D. 2.61 cm. Is the sample from a large population of mean 3.25 cm and S.D. of 2.61 cm? (Test at 5% level of significance.)

Soln :-

Null Hypothesis :- (H_0) The sample has been drawn from the population with mean $\mu = 3.25$ cm and S.D = 2.61 cm

Alternative Hypothesis (H_1) :- The Sample has not been from the given population ($\mu \neq 3.25$ cm)

It represents a two tailed test.

Test Statistic $z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

Here $\bar{x} = 3.4$; $\mu = 3.25$; $\sigma = 2.61$; $n = 900$.

$$\therefore z = \frac{3.4 - 3.25}{\left(\frac{2.61}{30}\right)} = 1.73$$

From the following table $z_\alpha = 1.96$

Hence $|z| < |z_\alpha|$, Accept the null Hypothesis.

An insurance agent has claimed that the average age of policy holders who insure through him is less than the average for all agents which is 30.5 years. A random sample of 100 policy holders who had insured through him reveal that the mean and S.D. are 28.8 years and 6.35 years respectively. Test his claim at 5% level of significance.

Soln:-

Null Hypothesis (H_0):- The average age of policy holders to the insurance agent is 30.5 years.

Alternative Hypothesis (H_1): The average age of policy holders to the insurance agent is less than 30.5 years.

This hypothesis represents a left tailed test.

Test Statistic $z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

Here $\bar{x} = 28.8$; $\mu = 30.5$ (σ is not given so we take that sample standard deviations)

$s = 6.35$; $n = 100$

$$\therefore z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{28.8 - 30.5}{6.35/10} = \frac{-17}{6.35}$$

$$= -2.6771.$$

From the tabular column

$$z_\alpha = -1.645$$

$\therefore |z| > |z_\alpha| \Rightarrow$ Reject the null Hypothesis.

A sample of 100 students is taken from a large population. The mean height of the students in this sample is 160 cm. Can it be reasonably regarded that, in the population, the mean height is 165 cm and S.D. is 10 cm? LOS is 1%.

Soln:-

Null Hypothesis (H_0) :- The difference between the population mean height and sample mean height are not significant ($\bar{x} = \mu$)

Alternative Hypothesis (H_1): $\bar{x} \neq \mu$.

The Hypothesis gives a two tailed test.

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Here $\bar{x} = 160$; $\mu = 165$

$\sigma = 10$; $n = 100$

$$\therefore Z = \frac{160 - 165}{10/10} = -5$$

From the tabular column

$Z_\alpha = 2.58$

$\therefore |Z| > |Z_\alpha|$ Reject Null Hypothesis.